

ABSTRACT

The University Campus Network Project aims to modernize the network infrastructure at ABC University by upgrading and redesigning the existing system. The project addresses issues such as unreliable connectivity, inadequate security measures, and limited scalability. Key aspects of the project include network segmentation, access control, enhanced security measures, wireless connectivity, network services, redundancy, and high availability. Documentation, monitoring, and reporting are integral components to ensure efficient management and proactive issue resolution. The project's objective is to optimize network performance, promote secure communication and collaboration, and provide seamless access to resources for students, faculty, and staff, thereby enhancing the overall network experience at the university.

1. Introduction:

This University Campus Network Scenario is about designing a network for a university in which various computers of different departments are set up so that they can interact and communicate with each other by interchanging data. To design a networking scenario for a university which connect various departments to each other's, it puts forward communication among different departments. Cisco Packet Tracer is used to design a systematic and well- planned network, satisfying all the necessities of the University Campus. Cisco Packet Tracer come up with a network with good performance. There are **Main Block, Admin Block, Engineering Block, Sir Syed Block, Sameed Block, Huzaifa Block & Anas Block** that can have access to have a communication.

2. Problem Statement:

The current network infrastructure at ABC University is outdated and inadequate to meet the growing demands of the campus community. The network suffers from various issues, including unreliable connectivity, limited security measures, and insufficient capability. These problems hinder effective communication, collaboration, and access to resources for students, faculty, and staff. Therefore, there is a need to revamp and upgrade the university's network infrastructure to address these challenges and provide a robust, secure, and scalable network that meets the evolving needs of the campus community. The goal is to ensure reliable connectivity, enhanced network security, efficient traffic flow, and seamless access to network services across the campus.

3. Scope of Project:

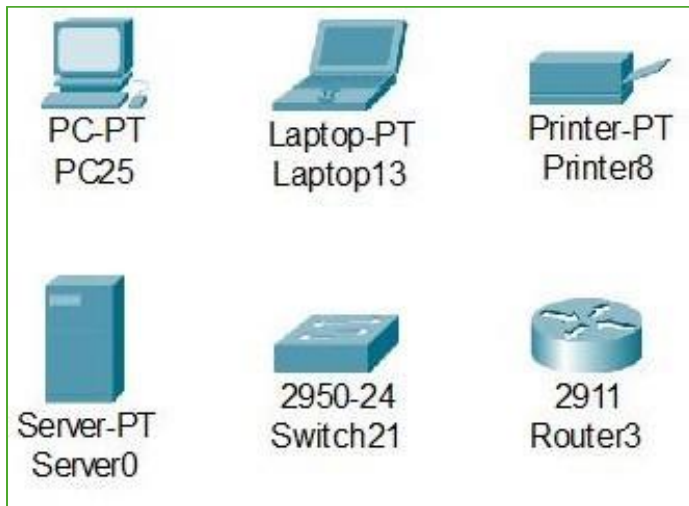
The scope of the project includes designing and configuring the network infrastructure, implementing network security measures, ensuring redundancy and high availability, setting up network services, documenting the network design and configurations, conducting testing and troubleshooting, and considering scalability and future expansion. The project aims to establish a reliable, secure, and scalable campus network that supports seamless connectivity, network segmentation, user

authentication, centralized services, and proactive monitoring. The scope may vary based on specific requirements and available resources.

4. Aims and Objectives:

The main objective of the proposed network is to update the existing network and also enhance its capabilities and increase the flexibility of the network which will eventually provide good security.

5. Devices Used:



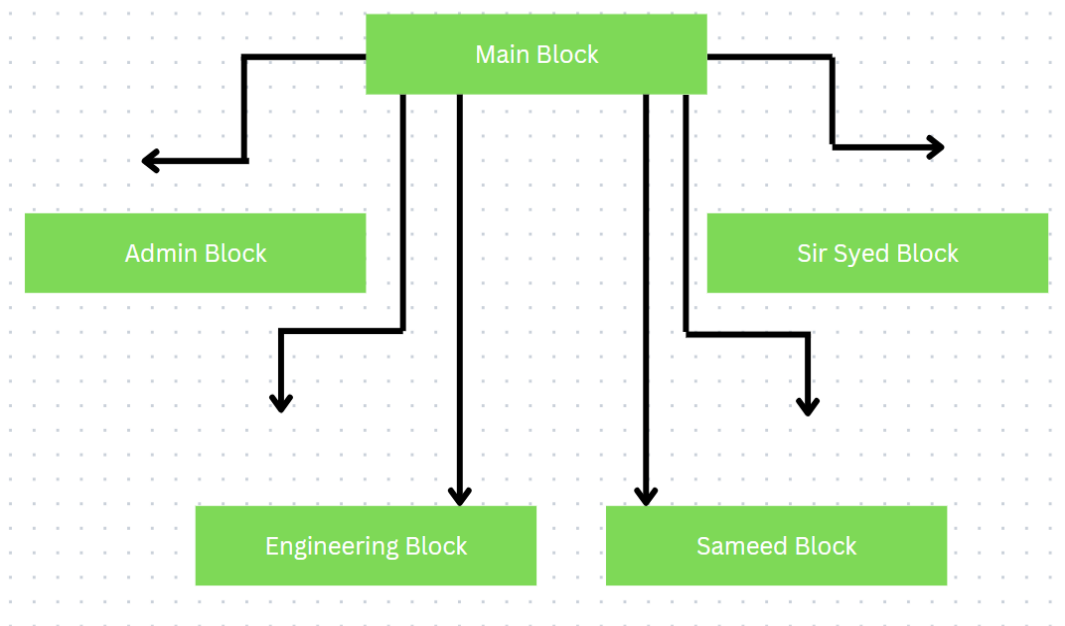
6. Functionality:

1. VLAN connectivity
2. ACL is applied.
3. Routers connectivity through RIP Trunking Port Enable.
4. Dynamic and static IP address is Applied.
5. Port Security.
6. Permanent Mac Address is applied.
7. Protected/ Secret Password.

7. Working:

Main Block have two router which is then divided into three rooms have two switches. First switch is having three room that is HOD, Teachers and Advisor. Second switch is having two switch which is then divided into two room that is Principal, Vice Principal and printer. And second router have one switch that is

connected which server to provide email services. We have given IPs to them dynamically and they are in VLAN 1 by default. ACL is applied on the second router by which student advisor can't communicate with email server and principal. Port is secured in email server and PCs of the principal is static. Router of the main block is then connected to the router of **Web Server**. Secret and Privileged mode password is applied to web server. Server is then attached to **Admin** Block and given IPs to the PCs dynamically and then it is dub divided into admin office and reception and given port security and static IPs. **Engineering** Block is then divided into CS and IT department, and they can communicate with each other. Port Security and permanent Mac address are applied to them. **Sir Syed** Block have two department EE and CE. Protective password is applied to them. **Anas** Block have two departments that is English and BBA. Port security and protective password are applied to them. Similar is the case for **Sameed** block and **Huzaiifa** block. Every Block can communicate with each other except for main block.

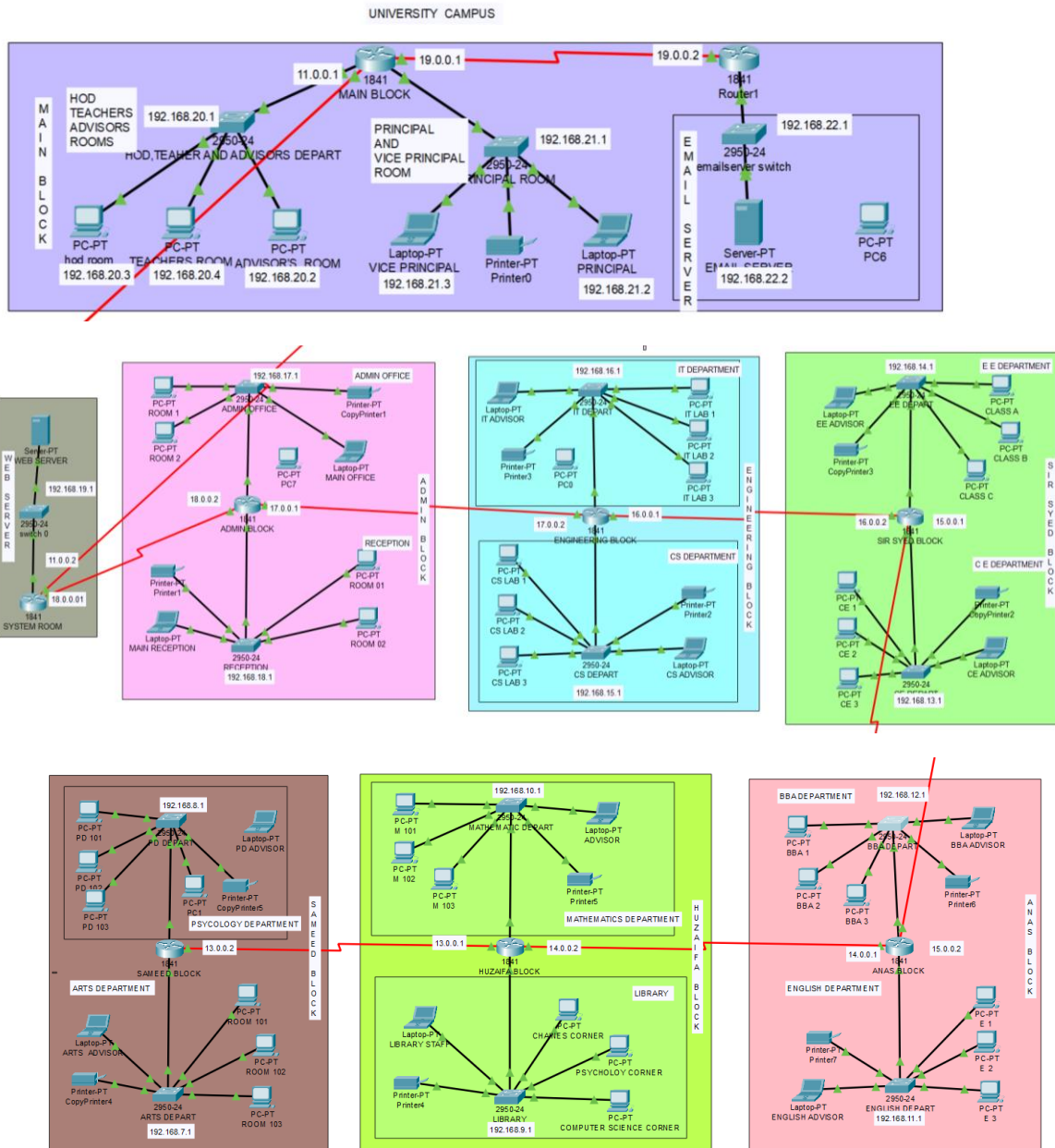


8. Tools:

- **Cisco Packet Tracer:**

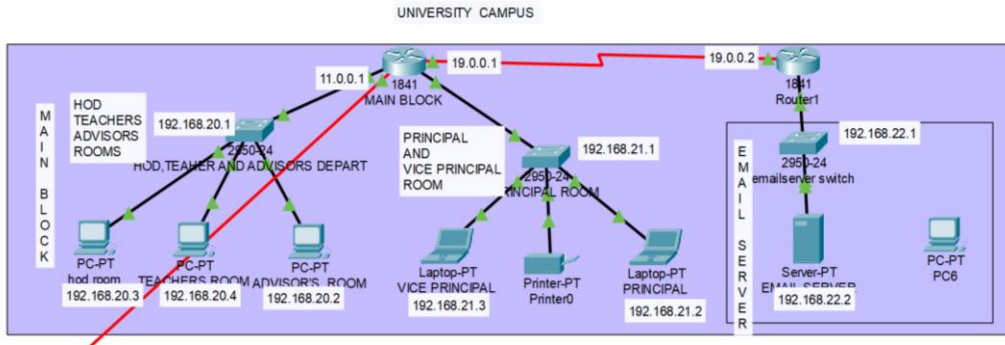
Cisco Packet Tracer is a network simulation and visualization tool designed by Cisco Systems. It is primarily used for educational and training purposes to simulate computer networks and their associated devices. Packet Tracer allows users to create virtual networks, configure network devices, and test network configurations in a simulated environment. It provides a hands-on learning experience for students and professionals to understand and practice networking concepts, such as routing, switching, and network protocols. Packet Tracer is widely used in networking courses and certifications to enhance understanding and skills development in the field of computer networking.

PROJECT SCREENSHOTS



MAIN BLOCK

Main Block is divided into sub-department. One switch provides the HOD, Teacher and Adviser connection. Second switch provides the connectivity for Principal and Vice Principal. And the third switch provide the Email connectivity. Router are connected through RIP trunking mode



MAIN BLOCK

Physical Config CLI

IOS Command Line Interface

```

Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#
Router(config-if)#EX
Router(config)#EX
Router#
$SYS-5-CONFIG_I: Configured from console by console

Router#CONFIG T
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool HTA-pool
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 192.168.20.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp pool principal-pool
Router(dhcp-config)#network 192.168.21.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.21.1
Router(dhcp-config)#dns-server 192.168.21.1
Router(dhcp-config)#ex
  
```

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HOD'S ROOM

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.20.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 192.168.20.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address

Link Local Address FE80::2D0:58FF:FE63:B128

IPv6 Gateway

IPv6 DNS Server

PRINCIPAL

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IP Address 192.168.21.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.21.1

DNS Server 192.168.21.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address

Link Local Address FE80::290:21FF:FEE9:99E7

IPv6 Gateway

IPv6 DNS Server

MAIN BLOCK

Physical Config CLI

RIP Routing

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Serial0/0/0

Serial0/0/1

Network Address

19.0.0.0

192.168.20.0

192.168.21.0

192.168.22.0

Remove

Equivalent IOS Commands

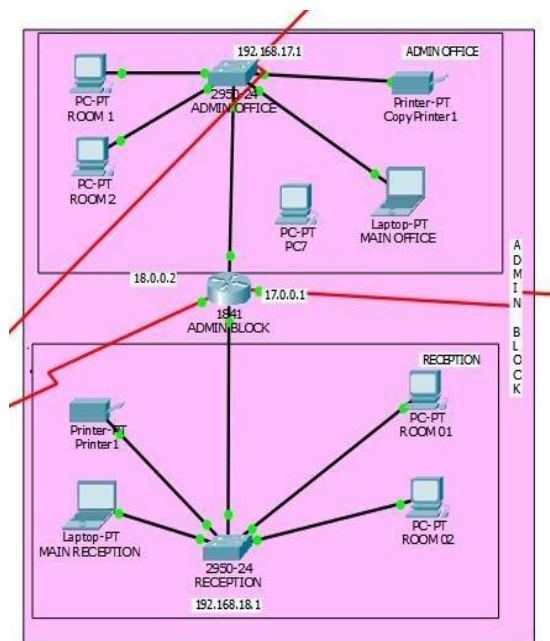
```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 19.0.0.0
Router(config-router)#network 192.168.20.0
Router(config-router)#network 192.168.21.0
Router(config-router)#network 192.168.22.0
Router(config-router)#
  
```

BECAUSE OF ACL APPLIED:

Scenario 0	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
New	Deleted	Failed	PRINC...	ADVISOR'S ...	ICMP		0.000	N	0

ADMIN BLOCK



ADMIN BLOCK

Physical Config CLI

IOS Command Line Interface

```

Router(config)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#ip dhcp pool admin-pool
Router(dhcp-config)#network 192.168.18.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.18.1
Router(dhcp-config)#dns-server 192.168.18.1
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp pool reception-pool
Router(dhcp-config)#network 192.168.17.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.17.1
Router(dhcp-config)#dns-server 192.168.17.1
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
  
```

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RIP ROUTING

ADMIN BLOCK

Physical Config CLI

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Serial0/0/0

Serial0/0/1

RIP Routing

Network

Add

Network Address

15.0.0.0

16.0.0.0

17.0.0.0

18.0.0.0

192.168.11.0

192.168.12.0

Remove

Equivalent IOS Commands

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%DHCeD-4-PING_CONFLICT: DHCP address conflict: server pinged 192.168.18.1.
%DHCeD-4-PING_CONFLICT: DHCP address conflict: server pinged 192.168.17.1.

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down
  
```

MAIN OFFICE:

ADMIN OFFICE

PhysicalConfigCLI

IOS Command Line Interface

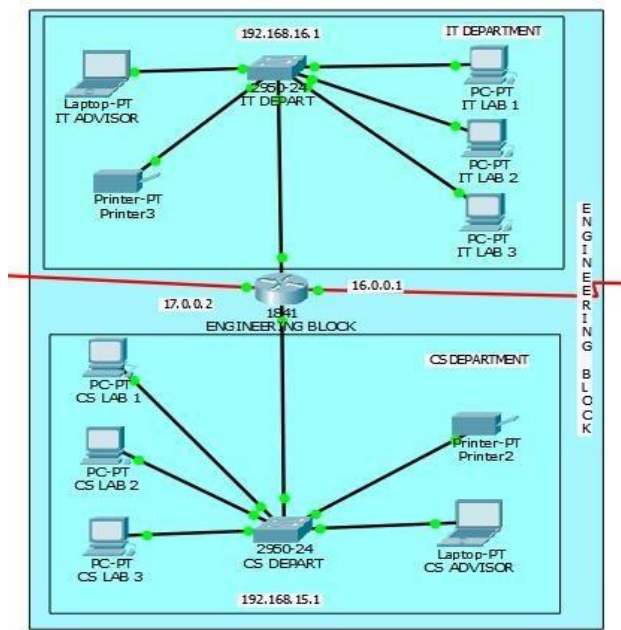
```
% Invalid input detected at '^' marker.

Switch(config)#INT FA 0/5
Switch(config-if)#SWITCHPORT MODE ACCESS
Switch(config-if)#SWITCHPORT PORT-SECURITY
Switch(config-if)#SWITCHPORT PORT-SECURITY MAXIMUM 1
Switch(config-if)#SWITCHPORT PORT-SECURITY MAC-ADDRESS STICKY
Switch(config-if)#SWITCHPORT PORT-SECURITY VIOLATION SHUTDOWN
Switch(config-if)#EX
Switch(config)#EX
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#SH MAC ADDRESS
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
-----
1       0001.642d.9973   STATIC    Fa0/5
1       0009.7cb4.9a02   DYNAMIC    Fa0/1
Switch#SH PORT-SECURITY
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/5      1              1              0      Shutdown
Switch#
```

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ENGINEERING BLOCK



ENGINEERING BLOCK

Physical Config CLI

IOS Command Line Interface

```

Router(config)#
$LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#ex
Router(config)#ip dhcp pool
$LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
down
$LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
% Incomplete command.
Router(config)#IP DHCP POOL ITDEPART-POOL
Router(dhcp-config)#network 192.168.16.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.16.1
Router(dhcp-config)#dns-server 192.168.16.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp pool csdepart-pool
Router(dhcp-config)#network 192.168.15.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.15.1
Router(dhcp-config)#dns-server 192.168.15.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
  
```

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CS ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

☒ DHCP
 ☐ Static

IP Address 192.168.15.3
 Subnet Mask 255.255.255.0
 Default Gateway 192.168.15.1
 DNS Server 192.168.15.1

IPv6 Configuration
☐ DHCP
 ☐ Auto Config
 ☒ Static

IPv6 Address
 Link Local Address FE80::201:42FF:FE6C:4D0D
 IPv6 Gateway
 IPv6 DNS Server

IT ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

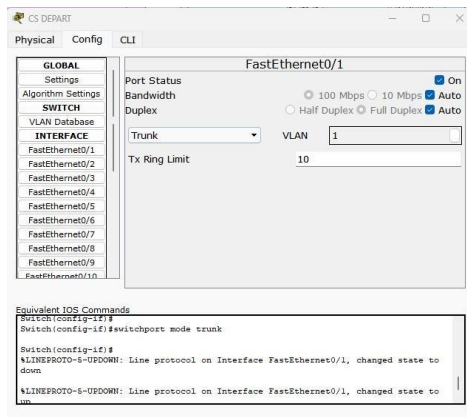
☒ DHCP
 ☐ Static

IP Address 192.168.16.3
 Subnet Mask 255.255.255.0
 Default Gateway 192.168.16.1
 DNS Server 192.168.16.1

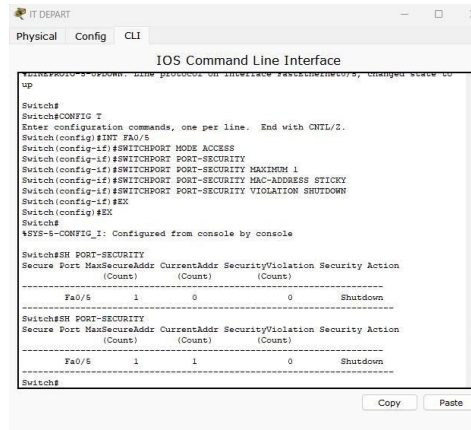
IPv6 Configuration
☐ DHCP
 ☐ Auto Config
 ☒ Static

IPv6 Address
 Link Local Address FE80::204:9AFF:FE39:8A37
 IPv6 Gateway
 IPv6 DNS Server

TRUNKING PORT:

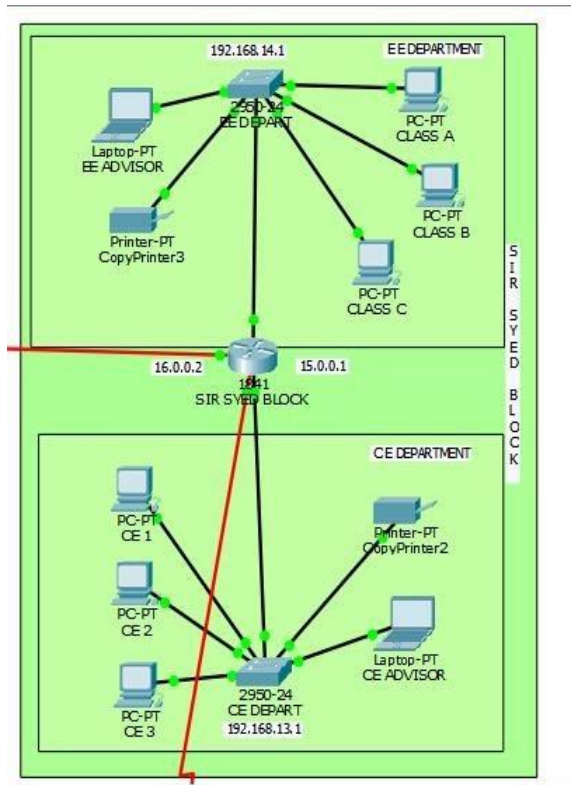


PORT SECURE: IT ADVISOR



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Failed	PC0	IT LAB 1	ICMP		0.000	N	0

SIR SYED BLOCK



Password Protected Department:



EE DEPART

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#ex
Switch(config-if)#line console 0
Switch(config-line)#password sara and ESHA
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#exit
```

Copy Paste



EE DEPART

Physical Config CLI

IOS Command Line Interface

```
User Access Verification

Password:

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable password sara and ESHA
Switch(config)#disable
Switch(config)#
Switch(config)#
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#exit
```

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EE DEPART

Physical Config CLI

IOS Command Line Interface

```
Press RETURN to get started.

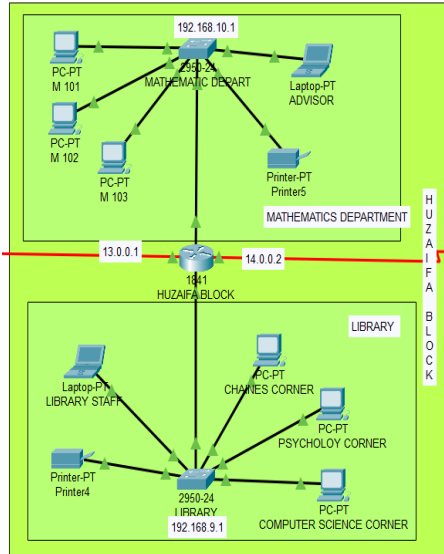
User Access Verification

Password:

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

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HUZAIFA BLOCK



Physical Config CLI

IOS Command Line Interface

```

Router(config)#interface FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 192.168.9.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#EX
Router(config-if)#IP DHCP POOL MATHDEPART-POOL
Router(dhcp-config)#NETWORK 192.168.10.0 255.255.255.0
Router(dhcp-config)#DEFAULT-ROUTER 192.168.10.1
Router(dhcp-config)#DNS-SERVER 192.168.10.1
Router(dhcp-config)#EX
Router(config)#DO WR
Building configuration...
[OK]
Router(config)#IP DHCP POOL LIBRARY-POOL
Router(dhcp-config)#NETWORK 192.168.9.0 255.255.255.0
Router(dhcp-config)#DEFAULT-ROUTER 192.168.9.1
Router(dhcp-config)#DNS-SERVER 192.168.9.1
Router(dhcp-config)#EX
Router(config)#DO WR
Building configuration...
[OK]
Router(config)#
  
```

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ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.10.5

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.10.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FE7A:91A5

IPv6 Gateway

IPv6 DNS Server

LIBRARY STAFF

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.9.5

Subnet Mask 255.255.255.0

Default Gateway 192.168.9.1

DNS Server 192.168.9.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

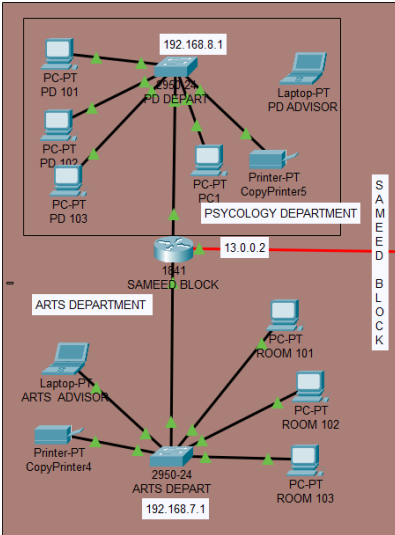
IPv6 Address /

Link Local Address FE80::2E0:F9FF:FEE1:70EE

IPv6 Gateway

IPv6 DNS Server

SAMEED BLOCK



```

Physical Config CLI
IOS Command Line Interface
Router(config)#interface FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 192.168.7.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#EX
Router(config)#IP DHCP POOL PPDEPART-POOL
Router(dhcp-config)#NETWORK 192.168.8.0 255.255.255.0
Router(dhcp-config)#DEFAULT-ROUTER 192.168.8.1
Router(dhcp-config)#DNS-SERVER 192.168.8.1
Router(dhcp-config)#EX
Router(config)#DO WR
Building configuration...
[OK]
Router(config)#IP DHCP POOL MARIDEPART-POOL
Router(dhcp-config)#NETWORK 192.168.7.0 255.255.255.0
Router(dhcp-config)#DEFAULT-ROUTER 192.168.7.1
Router(dhcp-config)#DNS-SERVER 192.168.7.1
Router(dhcp-config)#EX
Router(config)#DO WR
Building configuration...
[OK]
Router(config)#
Copy Paste

```

PP ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address: 192.168.8.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.8.1

DNS Server: 192.168.8.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::2D0:97FF:FE02:17B8

IPv6 Gateway: /

IPv6 DNS Server: /

MARI ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address: 192.168.7.6

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.7.1

DNS Server: 192.168.7.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:64FF:FE41:CA96

IPv6 Gateway: /

IPv6 DNS Server: /

Port Security for P ADVISOR:

```

Physical Config CLI
IOS Command Line Interface
1 0004.9a55.b358 DYNAMIC Fa0/3
1 0060.3e03.0e01 DYNAMIC Fa0/1
1 0060.70ea.b173 DYNAMIC Fa0/2
1 0040.9702.17b8 DYNAMIC Fa0/5
1 00e0.f753.a7c7 DYNAMIC Fa0/4
Switch(config)#
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#switchport mode access
% Invalid input detected at '^' marker.
Switch(config)#int fa 0/5
%LINK-S-CHANGED: Interface FastEthernet0/5, changed state to down
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINK-S-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch(config-if)#int fa 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#
Copy Paste

```

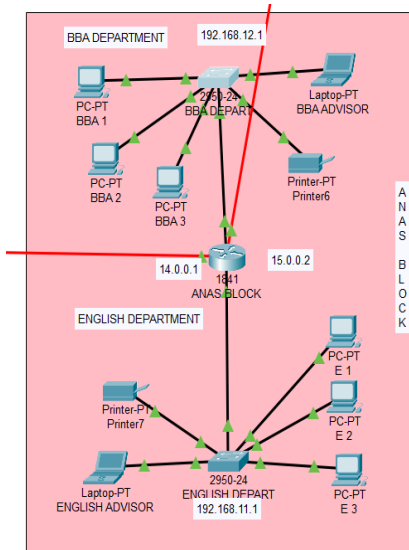
```

Physical Config CLI
IOS Command Line Interface
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#ex
Switch(config)#sh mac-address-table
% Invalid input detected at '^' marker.
Switch(config)#ex
Switch#
%SYS-S-CONFIG_I: Configured from console by console
Switch#sh mac-address-table
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0060.3e03.0e01    DYNAMIC   Fa0/1
Switch#sh mac-address-table
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0060.70ea.b173    DYNAMIC   Fa0/2
1     0040.9702.17b8    STATIC    Fa0/5
1     00e0.f753.a7c7    DYNAMIC   Fa0/4
Switch#
Copy Paste

```


Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Failed	PC1	PP 102	ICMP		0.000	N	0

ANAS BLOCK



QUAID BLOCK

Physical Config CLI

IOS Command Line Interface

```

Router(config)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#ip dhcp pool BBADEPART-POOL
Router(dhcp-config)#network 192.168.12.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.12.1
Router(dhcp-config)#dns-server 192.168.12.1
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp pool englishDEPART-POOL
Router(dhcp-config)#network 192.168.11.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.11.1
Router(dhcp-config)#dns-server 192.168.11.1
Router(dhcp-config)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

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BBA ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.12.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.12.1

DNS Server 192.168.12.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address

Link Local Address FE80::2E0:B0FF:FE92:D625

IPv6 Gateway

IPv6 DNS Server

ENGLISH ADVISOR

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 192.168.11.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.11.1

DNS Server 192.168.11.1

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

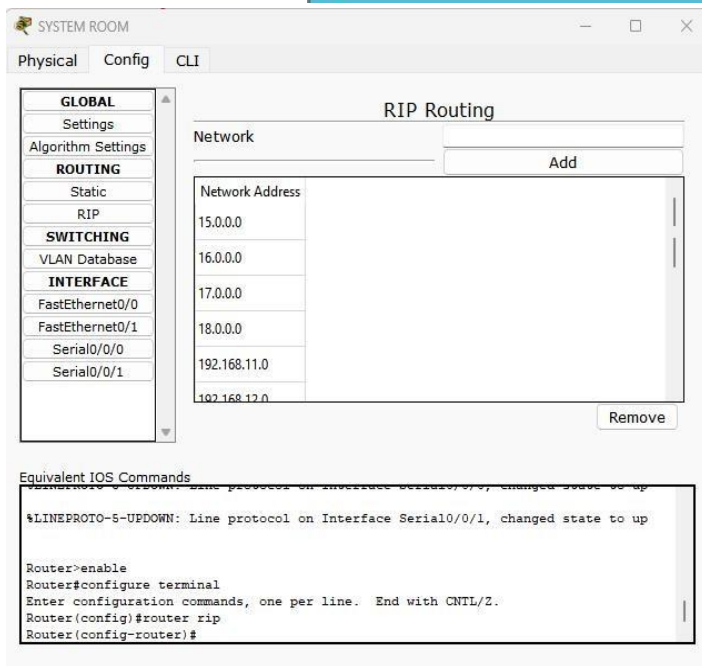
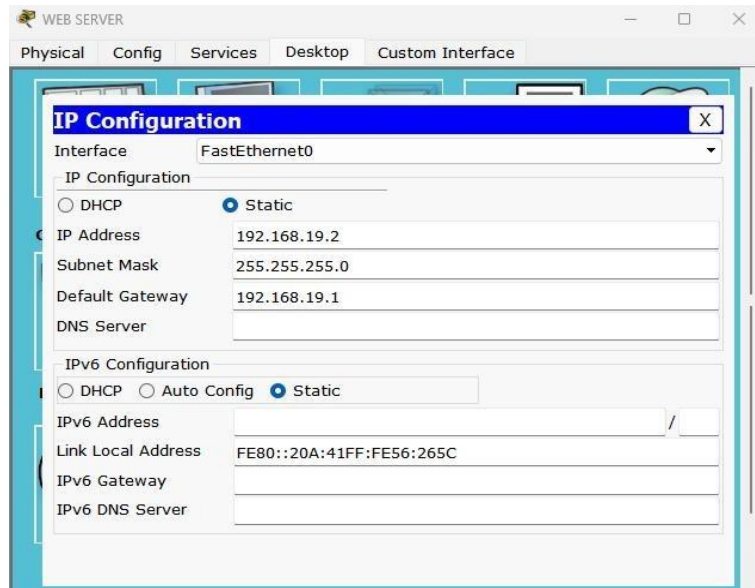
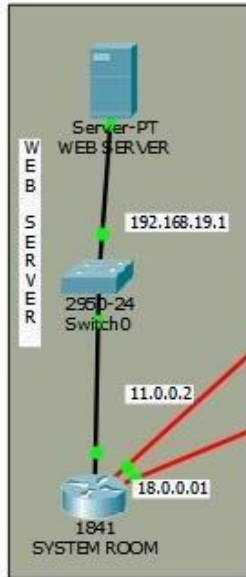
IPv6 Address

Link Local Address FE80::202:16FF:FE74:426

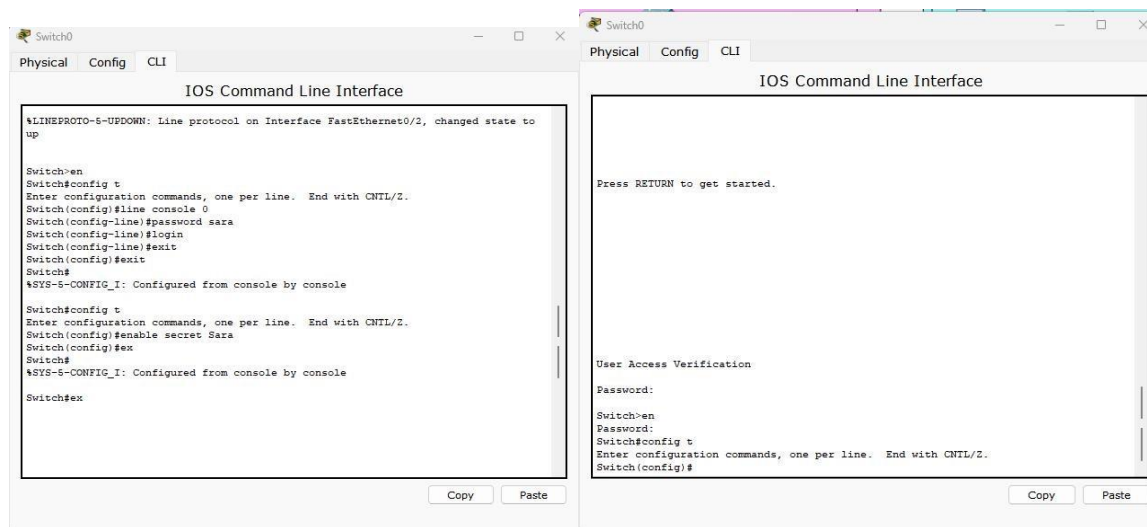
IPv6 Gateway

IPv6 DNS Server

SYSTEM ROOM:



Password Protective:



9. Conclusion:

The designed University Campus network in Cisco Packet Tracer provides a robust, secure, and scalable infrastructure to meet the networking needs of ABC University. It ensures reliable connectivity, efficient traffic flow, and enhanced network security. The network design promotes effective communication and collaboration across various departments and offers a seamless user experience for students, faculty, and staff.